

Online Retail II Dataset — Business Summary & Conclusions

Analysis of transactions from December 2009 to December 2011

Executive Summary

The dataset contains **1,067,371** transaction lines covering **01 Dec 2009** to **09 Dec 2011**. It spans **43** countries, about **5,942** identified customers, **5,305** product codes and **53,628** invoice numbers. Using Quantity × Price, the dataset's net transaction value is approximately **£19,287,251**. Positive-sales lines contribute **£20,814,292**, while negative-quantity lines represent returns/cancellations of about **£1,527,041**.

Metric	Value
Transaction lines	1,067,371
Invoice numbers	53,628
Identified customers	5,942
Products / Stock Codes	5,305
Countries	43
Positive-sales value	£20,814,292
Returns / negative value	-£1,527,041
Net transaction value	£19,287,251
Average positive invoice value	£496.24

Year-by-Year Performance

Period	Net value	Positive sales	Invoices	Customers
2009-2010	£9,539,485	£10,169,714	28,816	4,383
2010-2011	£9,747,766	£10,644,578	25,900	4,372

Positive-sales value increased from approximately £10.17M in 2009–2010 to £10.64M in 2010–2011, an increase of about 4.7%. This indicates moderate underlying growth, while the number of invoices declined, suggesting that sales value was supported by larger or more valuable orders.

Geographic Insights

Country	Positive sales
United Kingdom	£17,712,675
EIRE	£664,432
Netherlands	£554,232
Germany	£431,262
France	£356,945
Australia	£169,968
Spain	£109,179
Switzerland	£101,011

The United Kingdom dominates the dataset with approximately £17,712,675 in positive sales, far ahead of other countries. This concentration means UK customer behavior is likely to have the largest impact on overall business performance.

Product & Customer Insights

- **REGENCY CAKESTAND 3 TIER** (22423) — approximately £344,563 positive sales.
- **Manual** (M) — approximately £341,090 positive sales.
- **DOTCOM POSTAGE** (DOT) — approximately £322,657 positive sales.
- **WHITE HANGING HEART T-LIGHT HOLDER** (85123A) — approximately £262,931 positive sales.
- **PAPER CRAFT , LITTLE BIRDIE** (23843) — approximately £168,470 positive sales.

Customer purchasing is relatively repeat-oriented: about **72.4%** of identified customers placed more than one positive-value invoice, and repeat customers averaged roughly **6.3** invoices. However, **243,007** rows (22.8%) have no Customer ID, which limits customer-level analysis and retention modeling.

Returns, Cancellations & Data Quality

There are **22,950** negative-quantity transaction lines. Their value is approximately **£1,527,041**, equivalent to about **7.3%** of positive sales value. These records should be retained for net-sales reporting but separated from standard sales KPIs such as average order value and product demand. The dataset also contains missing Customer IDs and special/non-product codes such as manual/postage entries, so data cleaning is important before predictive modeling.

Key Conclusions

- Overall performance is positive: the dataset records more than £20.8M in positive sales value across the two periods.
- Sales value grew modestly in the second period, despite fewer invoices, pointing to an improvement in value per invoice/order.
- The UK is the core market, so geographic diversification could reduce dependence on a single market.
- Repeat purchasing is meaningful, making customer retention, segmentation and loyalty analysis promising areas for further work.
- Returns/cancellations are material and should be monitored by product, customer and month to identify avoidable revenue leakage.
- Data quality needs attention: missing Customer IDs and special transaction codes can distort customer and product analysis.
- Seasonality appears important, with November showing the strongest monthly positive sales in both observed years; promotional and holiday planning should account for this pattern.

Recommended Next Analysis

- Build an RFM (Recency, Frequency, Monetary) customer segmentation using only valid customer records.
- Create a monthly sales dashboard showing gross sales, returns, net sales, invoices, average order value and customer count.
- Analyze return rates by product and country to identify operational or product-quality issues.
- Identify top customers and their contribution to revenue, while checking concentration risk.
- Separate regular product sales from postage, manual adjustments and other non-standard codes before product-level modeling.

Method note: Revenue/value is calculated as Quantity × Price. Negative quantities are treated as returns/cancellations. Figures are descriptive and based solely on the uploaded Online Retail II workbook.